Ysbyty Gwynedd
Anaesthetics Department
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PERIOPERATIVE & POST-OPERATIVE MANAGEMENT

Peri-Operative Management of Emergency Laparotomies

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Key Points

- 30-day mortality rates is ~10% in the UK.
- Document a Formalised Risk assessment.
- Direct Consultant involvement for high-risk patients.
- Goal Directed Fluid Therapy.
- Admit high-risk patients to critical care.
- Document a comprehensive post op plan.
- Complete On-line NELA Database (Section 1-6)

Useful Number: Aseptic Department: 4137

HIGH RISK INDICATORS
NELA Risk Score ≥ 5%
Clinical Frailty Score ≥4
ASA 3-5
Insulin Dependent Diabetes
Acute Kidney Injury or CKD ≥3b (eGFR <45)
Sepsis & Lactate >2mol/L
Immunosupressive Therapy

Key elements of peri-operative care

1. **Prompt identification of patients most at risk of deterioration based on their NEWS score, Lactate & assessment of frailty**
2. **Consultant delivered care throughout the perioperative journey for high risk patients**
3. **Considered intra-operative anaesthetic approach**
4. **Formalised post-operative management plan**

Identification of patients most at risk

- A formalised mortality risk assessment is an integral component of preoperative assessment. This is based on physiological parameters, risk prediction tools (NELA), frailty assessments (Clinical Frailty Score, AWOL) and clinical judgment.
- Assessment should include review of antimicrobial cover, acid-base status and albumin levels.
- Document whether low or high risk, NELA prediction and Clinical Frailty Score on the anaesthetic record.

Consultant delivered care

- The consultant anaesthetist must be informed of any high risk patients. It is deemed best practice for the consultant anaesthetist to undertake a preoperative review for all patients at high risk.
- Direct consultant presence throughout the intra-operative period, in particular for induction and emergence, as is considered best practice for high risk patients.

Intra-Operative management

- **Avoid Hypotension**
  - Ensure cardiac stable induction.
  - Low threshold for invasive BP monitoring.
- **Optimise Lung Function**
  - Lung Protection Ventilation
  - Deep muscle relaxation with objective NMT monitoring (TOF≥0.9 pre extubation).
  - Optimise lung recruitment prior to and after extubation, particularly in obese patients.
- **Goal Directed Fluid Therapy (GDFT)**
  - Appropriate dynamic fluid resuscitation & optimisation using pulse pressure variation (PPV) or oesophageal Doppler (ODM+).
- Use ROTEM based blood therapy in suspected coagulopathy.
- Hypomagnesaemia is common. Consider administration of 40-50mg/kg.
- **Opioid Sparing Analgesia** (See Figure 1)

Post-Operative management plan

- Reassess the mortality risk after operative interventions using physiological parameters, risk prediction tools (NELA), frailty assessments and clinical judgment.
- Admit all high-risk patients to critical care.
- Consider extended recovery.
- Share and agree on post-operative plan with the surgical team at WHO Sign-Out.
- Prescribe and document on anaesthetic record the following: Target MAP, target SP02, fluid rate, analgesia, antibiotic therapy, DVT prophylaxis, nutritional instructions.
- Inform the surgical team if a Care of the Elderly Physician review is considered appropriate.

Documentation

- Complete NELA form on the NELA website (Sections 1-6).
- Document post-operative risk assessment (high or low risk)
- Document post-operative plan as described above.
- Complete Acute Pain Team referral form.

Appropriateness of surgery: MDT Discussion

- When the predicted in-hospital mortality is ≥25%, by any measure including frailty or clinical concerns exist regarding appropriateness of proceeding with surgery, they should be assessed pre-operatively in person by a consultant surgeon and consultant anaesthetist/intensivist to provide multidisciplinary advice to the patient about operative and non-operative treatment options and to discuss postoperative ceilings of care.
- Follow shared decision making principles.
- A joint discussion is undertaken between the surgical, anaesthesia and critical care consultants, followed by a joint meeting with the patient and relatives.
- Advice may be sought from the palliative care team:
  - 09:00-17:00: Bleep 183 Out of Hours advice line: 01978 316800

Links to other relevant guidelines

- YG CG 011: Rectus Sheath Guidelines
- YG CG Emergency Laparotomy Pathway

References

- www.nela.org.uk
- RCS Report The HighRisk General Surgical Patient Raising the Standard December 2018
- Link to Rockwood Clinical Frailty Score: www.camapcanada.ca/Frailtysale.pdf
- Alveolar recruitment maneuver and perioperative ventilatory support in obese patients undergoing abdominal surgery. Rev Bras Ter Intensiva. 2013 Oct-Dec; 25(4): 312–318.

Figure 1: Adult Emergency Laparotomy Analgesia Guide

